3. Show that the three updated probabilities sum to 1. (Carry each probability to four decimal places.)

   Solution:

   The sum of the three updated probabilities is

   $$P(\textit{EPS exceeded consensus}|\textit{DriveMed expands}) + P(\textit{EPS met consensus}|\textit{DriveMed expands}) + P(\textit{EPS fell short of consensus}|\textit{DriveMed expands})$$

   $$= 0.8232 + 0.1463 + 0.0305 = 1.000$$

   The three events (*EPS exceeded consensus*, *EPS met consensus*, and *EPS fell short of consensus*) are mutually exclusive and exhaustive: One of these events or statements must be true, so the conditional probabilities must sum to 1. Whether we are talking about conditional or unconditional probabilities, whenever we have a complete set of distinct possible events or outcomes, the probabilities must sum to 1. This calculation serves to check your work.

4. Suppose, because of lack of prior beliefs about whether DriveMed would meet consensus, you updated on the basis of prior probabilities that all three possibilities were equally likely: $P(\textit{EPS exceeded consensus}) = P(\textit{EPS met consensus}) = P(\textit{EPS fell short of consensus}) = 1/3$.

   Solution:

   Using the probabilities given in the question,

   $$P(\textit{DriveMed expands})$$

   $$= P(\textit{DriveMed expands}|\textit{EPS exceeded consensus})P(\textit{EPS exceeded consensus}) + P(\textit{DriveMed expands}|\textit{EPS met consensus})P(\textit{EPS met consensus}) + P(\textit{DriveMed expands}|\textit{EPS fell short of consensus})P(\textit{EPS fell short of consensus})$$

   $$= 0.75(1/3) + 0.20(1/3) + 0.05(1/3) = 1/3$$

   Not surprisingly, the probability of DriveMed expanding is one-third (1/3) because the decision maker has no prior beliefs or views regarding how well EPS performed relative to the consensus estimate.

   Now we can use Bayes' formula to find $P(\textit{EPS exceeded consensus} \mid \textit{DriveMed expands}) = [P(\textit{DriveMed expands} \mid \textit{EPS exceeded consensus})/P(\textit{DriveMed expands})]\ P(\textit{EPS exceeded consensus}) = [(0.75/(1/3)]\ (1/3) = 0.75$, or 75 percent. This probability is identical to your estimate of $P(\textit{DriveMed expands} \mid \textit{EPS exceeded consensus})$.

   When the prior probabilities are equal, the probability of information given an event equals the probability of the event given the information. When a decision maker has equal prior probabilities (called **diffuse priors**), the probability of an event is determined by the information.

QUESTION SET

The following example shows how Bayes' formula is used in credit granting in cases in which the probability of payment given credit information is higher than the probability of payment without the information.

1. Jake Bronson is predicting the probability that consumer finance applicants granted credit will repay in a timely manner (i.e., their accounts will not